

Exercise 1.8

Lagrangian:
$$L = \int_0^l dx \mathcal{L} = \int_0^l dx \left[\frac{\mu}{2} \left(\frac{\partial y}{\partial t} \right)^2 - \frac{T}{2} \left(\frac{\partial y}{\partial x} \right)^2 \right]$$

We express $y(x, t)$ as a fourier transform such that

$$y(x, t) = \sqrt{\frac{2}{l}} \sum_{n=1}^{\infty} \sin\left(\frac{n\pi x}{l}\right) q_n(t)$$

Plugging this into equation for L we get

$$\begin{aligned} L &= \int_0^l dx \left[\frac{\mu}{2} \left(\frac{\partial}{\partial t} \left[\sqrt{\frac{2}{l}} \sum_{n=1}^{\infty} \sin\left(\frac{n\pi x}{l}\right) q_n(t) \right] \right)^2 - \frac{T}{2} \left(\frac{\partial}{\partial x} [\dots] \right)^2 \right] \\ &= \int_0^l dx \left[\frac{\mu}{2} \cdot \frac{2}{l} \left(\sum_{n=1}^{\infty} \sin\left(\frac{n\pi x}{l}\right) \dot{q}_n(t) \right)^2 - \frac{T}{2} \cdot \frac{2}{l} \left(\sum_{n=1}^{\infty} \frac{n\pi}{l} \cos\left(\frac{n\pi x}{l}\right) q_n(t) \right)^2 \right] \end{aligned}$$

The cross terms in $\left(\sum_{n=1}^{\infty} \sin\left(\frac{n\pi x}{l}\right) \dot{q}_n(t) \right)^2$ and $\left(\sum_{n=1}^{\infty} \frac{n\pi}{l} \cos\left(\frac{n\pi x}{l}\right) q_n(t) \right)^2$

vanish when doing the integral since

$$\begin{aligned} \int_0^l dx \sin\left(\frac{n\pi x}{l}\right) \sin\left(\frac{k\pi x}{l}\right) &= \frac{l \sin(\pi k) \cos(\pi n) - k l \cos(\pi k) \sin(\pi n)}{\pi k^2 - \pi n^2} = 0 \\ - \int_0^l dx \cos\left(\frac{n\pi x}{l}\right) \cos\left(\frac{k\pi x}{l}\right) &= \frac{-k l \sin(\pi k) \cos(\pi n) + l n \cos(\pi k) \sin(\pi n)}{\pi k^2 - \pi n^2} = 0 \end{aligned}$$

These are zero since $\sin(\pi k) = 0$, $\sin(n\pi) = 0$ for $k, n \in \mathbb{Z}$

this means that under the integral we can write

$$\left(\sum_{n=1}^{\infty} \sin\left(\frac{n\pi x}{l}\right) \dot{q}_n(t) \right)^2 \text{ as } \sum_{n=1}^{\infty} \sin^2\left(\frac{n\pi x}{l}\right) \dot{q}_n^2(t)$$

and $\left(\sum_{n=1}^{\infty} n \cos\left(\frac{n\pi x}{l}\right) q_n(t) \right)^2 \text{ as } \sum_{n=1}^{\infty} n^2 \cos^2\left(\frac{n\pi x}{l}\right) q_n^2(t)$

So we have
$$L = \sum_{n=1}^{\infty} \int_0^l dx \left[\frac{\mu}{l} \sin^2\left(\frac{n\pi x}{l}\right) \dot{q}_n^2(t) - \frac{T}{l^3} n^2 \pi^2 \cos^2\left(\frac{n\pi x}{l}\right) q_n^2(t) \right]$$

Let's compute the integrals (via calculator since I'm too lazy)

$$\int_0^l dx \sin^2\left(\frac{\pi n x}{l}\right) = \frac{1}{4} l \left(2 - \frac{\sin(2\pi n)}{\pi n} \right) \quad \parallel \sin(2\pi n) = 0$$

$$= \frac{1}{2} l$$

$$\int_0^l dx \cos^2\left(\frac{\pi n x}{l}\right) = \frac{l}{4\pi n} (2\pi n + \sin(2\pi n)) \quad \parallel \sin(2\pi n) = 0$$

$$= \frac{l}{2}$$

Inputting the results into L we get

$$L = \sum_{n=1}^{\infty} \left[\frac{\mu}{2} \dot{q}_n(t)^2 - \frac{T}{2l^2} n^2 \pi^2 q_n(t)^2 \right] \quad \parallel \text{We define } \omega_n = \sqrt{\frac{T}{\mu}} \frac{n\pi}{l}$$

$$\Rightarrow L = \sum_{n=1}^{\infty} \left[\frac{1}{2} \mu \dot{q}_n^2 - \frac{1}{2} \mu \omega_n^2 q_n^2 \right] \quad \square$$

We have reached the desired form for the Lagrangian.

Exercise 1.9

$$\text{Hamiltonian density: } \mathcal{H} = \pi^a \dot{y}_a - \mathcal{L} \quad \parallel \pi^a = \frac{\partial \mathcal{L}}{\partial \dot{y}_a}$$

$$\text{And Hamiltonian is } H = \int dx \mathcal{H}$$

We are working on one-dimensional problem:

$$H = \int dx \mathcal{H} = \int_0^l dl (\pi_y \dot{y} - \mathcal{L})$$

$$= \left(\int_0^l dl \pi_y \dot{y} \right) - L$$

$$= \int_0^l dl \mu \dot{y}^2 - L$$

$$H = \left[\sum_{n=1}^{\infty} \int_0^l dl \frac{2\mu}{l} \sin^2\left(\frac{n\pi x}{l}\right) \dot{q}_n^2(t) \right] - L$$

$$H = \sum_{n=1}^{\infty} \frac{2\mu}{l} \dot{q}_n^2 \cdot \frac{1}{2} l - L$$

$$\dot{y} = \sqrt{\frac{2}{l}} \sum_{n=1}^{\infty} \sin\left(\frac{n\pi x}{l}\right) \dot{q}_n(t)$$

$$\pi_y = \frac{\partial \mathcal{L}}{\partial \dot{y}} = \frac{\partial}{\partial \dot{y}} \left(\frac{\mu}{2} \dot{y}^2 \right) = \mu \dot{y}$$

using same arguments as in previous exercise we simplify $\dot{y}^2$ term inside integral

We use the solution to integral from previous exercise

We now put the equation for L

$$\Rightarrow H = \sum_{n=1}^{\infty} \mu \dot{q}_n^2 - \sum_{n=1}^{\infty} \left[\frac{1}{2} \mu \dot{q}_n^2 - \frac{1}{2} \mu \omega_n^2 q_n^2 \right]$$

$$H = \sum_{n=1}^{\infty} \left[\frac{1}{2} \mu \dot{q}_n^2 + \frac{1}{2} \mu \omega_n^2 q_n^2 \right]$$

in terms of $p_n = \mu \dot{q}_n$

where p_n is the canonical momentum for n -th coordinate. we get:

$$H = \sum_{n=1}^{\infty} \left[\frac{1}{2\mu} p_n^2 + \frac{1}{2} \mu \omega_n^2 q_n^2 \right]$$

The canonical commutation relations are $[q_n, p_m] = i \delta_{nm}$

with other commutations being 0 i.e. $[q_n, q_m] = 0 = [p_n, p_m]$

This leads directly to $[y(x,t), y(x',t)] = 0$, since $y \sim q_n$ &
 $[\pi_y(x,t), \pi_y(x',t)] = 0$ $\pi_y \sim p_n$

Now let's calculate $[y(x,t), \pi_y(x',t)]$

$$y(x,t) = \sqrt{\frac{2}{\ell}} \sum_{n=1}^{\infty} \sin\left(\frac{n\pi x}{\ell}\right) q_n(t)$$

$$\pi_y(x',t) = \mu \dot{y}(x',t) = \sqrt{\frac{2}{\ell}} \sum_{n=1}^{\infty} \sin\left(\frac{n\pi x'}{\ell}\right) p_n(t)$$

$$\Rightarrow [y(x,t), \pi_y(x',t)] = \frac{2}{\ell} \left(\sum_{n=1}^{\infty} \sin\left(\frac{n\pi x}{\ell}\right) q_n \right) \left(\sum_{n=1}^{\infty} \sin\left(\frac{n\pi x'}{\ell}\right) p_n \right) - \frac{2}{\ell} \left(\sum_{n=1}^{\infty} \sin\left(\frac{n\pi x'}{\ell}\right) p_n \right) \left(\sum_{n=1}^{\infty} \sin\left(\frac{n\pi x}{\ell}\right) q_n \right)$$

$$= \frac{2}{\ell} \sum_{n=1}^{\infty} \sum_{k=1}^{\infty} \sin\left(\frac{n\pi x}{\ell}\right) \sin\left(\frac{k\pi x'}{\ell}\right) [q_n, p_k] = i \delta_{nk}$$

$$= \frac{2}{\ell} i \sum_{n=1}^{\infty} \sin\left(\frac{n\pi x}{\ell}\right) \sin\left(\frac{n\pi x'}{\ell}\right)$$

Using the equations $\begin{cases} \cos(x+y) = \cos(x)\cos(y) - \sin(x)\sin(y) \\ \cos(x-y) = \cos(x)\cos(y) + \sin(x)\sin(y) \end{cases}$

We recover equation for $\sin(x)\sin(y)$:

$$\sin(x)\sin(y) = \frac{1}{2} (\cos(x-y) - \cos(x+y))$$

$$\text{This leads to } [y(x,t), \pi_y(x',t)] = \frac{i}{\ell} \sum_{n=1}^{\infty} \cos\left(\frac{n\pi}{\ell} (x-x')\right) - \cos\left(\frac{n\pi}{\ell} (x+x')\right)$$

The sum for the case $n=0$ is zero so we can write

$$[y(x,t), \pi_y(x',t)] = \frac{i}{\ell} \sum_{n=0}^{\infty} \cos\left[\frac{n\pi}{\ell} (x-x')\right] - \cos\left[\frac{n\pi}{\ell} (x+x')\right] \quad \left\| \begin{array}{l} \text{Using equation} \\ \text{provided we simplify} \\ \text{this to:} \end{array} \right.$$

$$[y(x,t), \pi_y(x',t)] = i \delta(x-x')$$

So we have shown $[y(x,t), y(x',t)] = 0 = [\pi_y(x,t), \pi_y(x',t)]$ &

$$[y(x,t), \pi_y(x',t)] = i \delta(x-x')$$

Exercise 1.10

Lagrangian density:

$$\mathcal{L} = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi + \frac{1}{2} \kappa (\partial_\mu \phi \partial^\mu \phi)^2 - \frac{1}{2} m^2 \phi^2 - g \phi^3 + \lambda \phi^4 - \sigma \phi^6$$

(a) System in 3 space dimension $\Rightarrow$ 4D-spacetime

Action is dimensionless: $[d^4x] = -4 \Rightarrow [\mathcal{L}] = 4$

Going term by term:

$$[x^\mu] = -1 \Rightarrow [\partial_\mu] = [\partial^\mu] = -1 \Rightarrow [\partial_\mu \phi \partial^\mu \phi] = 2[\partial_\mu] + 2[\phi]$$

$$4 = -2 + 2[\phi]$$

$$\text{We know } [\partial_\mu \phi \partial^\mu \phi] = [\mathcal{L}] = 4$$

$$\Rightarrow 2[\phi] = 4 + 2 = 6$$

$$\Rightarrow \underline{\underline{[\phi] = 3}}$$

$$\text{we also have } [\kappa (\partial_\mu \phi \partial^\mu \phi)^2] = 4 = [\kappa] + 2[\partial_\mu \phi \partial^\mu \phi] = [\kappa] + 2 \cdot 4$$

$$\Rightarrow \underline{\underline{[\kappa] = -4}}$$

$$[m^2 \phi^2] = 4 \Rightarrow 2[m] + 2[\phi] = 4 \Rightarrow 2[m] = -2 \Rightarrow \underline{\underline{[m] = -1}}$$

$$[g \phi^3] = 4 \Rightarrow \underline{\underline{[g] = -5}}, \quad [\lambda \phi^4] = 4 \Rightarrow \underline{\underline{[\lambda] = -8}}$$

$$[\sigma \phi^6] = 4 \Rightarrow \underline{\underline{[\sigma] = -14}}$$

The mass dimensions are $[\phi] = 3, [\kappa] = -4, [m] = -1, [g] = -5, [\lambda] = -8$
and $[\sigma] = -14$ for 3-space dimensions

b) For 2 space dimensions we have 3D-spacetime

$$[S] = 0 \Rightarrow [\mathcal{L}] = 3 \text{ \& } [\partial_\mu] = -1 \text{ \& } [\partial_\mu \phi \partial^\mu \phi] = 3 \Rightarrow \underline{\underline{[\phi] = \frac{5}{2}}}$$

$$[\kappa (\partial_\mu \phi \partial^\mu \phi)^2] = [\kappa] + 2 \cdot 3 = 3 \Rightarrow \underline{\underline{[\kappa] = -3}}$$

$$[m^2 \phi^2] = 3 \Rightarrow 2[m] = 3 - 2[\phi] = -2 \Rightarrow \underline{\underline{[m] = -1}}$$

$$[g\phi^3] = 3 \Rightarrow [g] = 3 - 3 \cdot \frac{5}{2} = 3 - \frac{15}{2} = -\frac{9}{2} \Rightarrow \underline{\underline{[g] = -\frac{9}{2}}}$$

$$[\lambda\phi^4] = 3 \Rightarrow [\lambda] = 3 - 4 \cdot \frac{5}{2} = -7 \Rightarrow \underline{\underline{[\lambda] = -7}}$$

$$[\sigma\phi^6] = 3 \Rightarrow [\sigma] = 3 - 6 \cdot \frac{5}{2} = -12 \Rightarrow [\sigma] = -12$$

The mass dimensions are $[\phi] = \frac{5}{2}$, $[\kappa] = -3$, $[m] = -1$, $[g] = -\frac{9}{2}$,
 $[\lambda] = -7$ and $[\sigma] = -12$ for 2 space dimensions.

c) 5 spatial dimensions $\Rightarrow$ 6D-spacetime

$$[S] = 0 \Rightarrow [L] = 6, [\partial_\mu] = -1$$

$$[\partial_\mu \phi \partial^\mu \phi] = 6 \Rightarrow 2[\phi] = 8 \Rightarrow \underline{\underline{[\phi] = 4}}$$

$$[\kappa (\partial_\mu \phi \partial^\mu \phi)^2] = 6 \Rightarrow [\kappa] = 6 - 2 \cdot 8 = -10 \Rightarrow \underline{\underline{[\kappa] = -10}}$$

$$[m^2 \phi^2] = 6 \Rightarrow 2[m] = 6 - 8 = -2 \Rightarrow \underline{\underline{[m] = -1}}$$

$$[g\phi^3] = 6 \Rightarrow [g] = 6 - 3 \cdot 4 = -6 \Rightarrow \underline{\underline{[g] = -6}}$$

$$[\lambda\phi^4] = 6 \Rightarrow [\lambda] = 6 - 4 \cdot 4 = -10 \Rightarrow \underline{\underline{[\lambda] = -10}}$$

$$[\sigma\phi^6] = 6 \Rightarrow [\sigma] = 6 - 6 \cdot 4 = -18 \Rightarrow \underline{\underline{[\sigma] = -18}}$$

The mass dimensions are $[\phi] = 4$, $[\kappa] = -10$, $[m] = -1$, $[g] = -6$,
 $[\lambda] = -10$, $[\sigma] = -18$ for 5 space dimensions